

Section 1.3: Perspectives and Issues in Machine Learning

Your Name

Your Institute

Perspectives in Machine Learning: Learning as Search

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- Example: Checkers learner
 - Hypothesis space = all evaluation functions formed by $w_0, \dots, w_6$
 - Learning = searching this continuous weight space

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 - Artificial neural networks

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- Understanding representation structure helps analyze learnability

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 - The structure of the hypothesis spaces they explore
- Useful for formal sample complexity and generalization bounds

Issues in Machine Learning: Core Questions (1)

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 - Particular hypothesis representations?
- How much training data is sufficient for reliable learning?

Issues in Machine Learning: Core Questions (2)

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- What bounds relate:
 - Confidence in learned hypotheses
 - Size of hypothesis space
 - Amount of training data?
- How can prior knowledge:
 - Guide generalization?
 - Help even when approximate or partially correct?
- What is the best strategy for selecting:
 - The next training experience?

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- What specific functions should the system attempt to learn?
- Can this decomposition be automated?
- How can the learner automatically:
 - Modify its representation
 - Improve its ability to represent and learn the target function?